Adapting PolSARproSim to Simulate Temporal Decorrelation in a Heterogeneous Mixed Forest

Razi Ahmed, Scott Hensley, Maxim Neumann, Marco Lavalle

Radar Science and Engineering
Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA

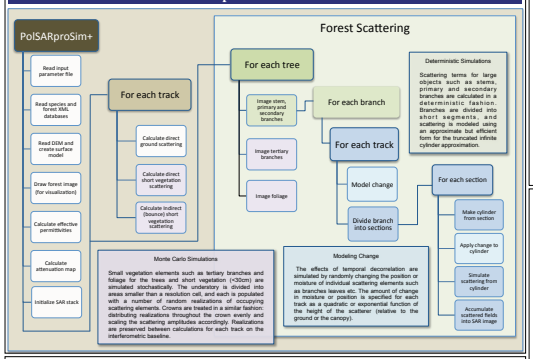
Paul Siqueira

Electrical And Computer Engineering
University of Massachusetts, Amherst, MA

Abstract

PolSARproSim [8] is a coherent forest SAR simulator, distributed as an open source software by the European Space Agency. Unlike other forest scattering softwares that model forests as multilayered homogenous structures with the canopies as clouds of randomly distributed scatterers, PolSARproSim simulates scattering from individual, intricately constructed trees with branches and leaves that have somewhat realistic shapes, dimensions, locations and dielectric properties. This feature is attractive because it allows an assessment of the impact of structure on forest parameter estimation techniques, such as polarimetric interferometry or tomography, in controlled yet somewhat realistic scenarios. Since PolSARproSim was written as an educational tool with ease of use in mind it has some limitations. For instance it can only simulate a homogenous stand, is limited to five species, generates tree parameters and locations based on some simple allometrics and does not introduce radar system artifacts such as thermal noise etc. In an effort to make the software more flexible and simulations more realistic, it has been modified to be able to simulate mixed stands, have external definition of species specific allometric parameters that essentially allow the user to simulate scattering from any number of tree species and to have the ability to specify tree by tree primitives such as location, height, dbh, crown radius and depth. In addition PolSARproSim has been modified to simulate temporal decorrelation based on pseudo-realistic changes in the physical properties of scattering elements inside a forest canopy. This is accomplished by introducing physics-based motion and dielectric change to individual tree elements such as branches and leaves.

PolSARproSim+ Process Flow



PolSARproSim+ Highlights

- PolSARproSim+ is intended to be run as a command line tool, written for Mac OSX, Windows and Linux platforms
- Simulation parameters such as scene geometry, radar parameters, forest and surface properties, moisture/motion profiles etc. can be set in a keyword-driven text-based parameter file
- Primitives (such as species, location, height, crown parameters, dbh etc) of each tree in a forest can be specified using an XML file. Species properties are controlled via a separate XML file.
- Any arbitrary DEM can be supplied using an external binary flat file. The DEM is resampled to simulation coordinates automatically. A shadow map is created based on the supplied DEM.
- An optical image of the forest can be generated for visualization.
- Effective permittivities for each species are calculated using a sample of the first ten trees of each species.
- A three dimensional attenuation map is created to approximate the attenuation experienced by each scatterer based on effective permittivities.
- For each track the direct scattering contribution from the ground is estimated using the SPM model and Monte Carlo techniques.
- A short layer (<30cm) of understory vegetation is simulated as homogenously distributed small needles and stems with both direct and double-bounce contributions using Monte Carlo techniques.
- Returns from each tree in the forest are simulated by dividing each branch into small sections and approximating each individual section as an oriented cylinder.
- By specifying coefficients of a vertical profile, the amount of motion or moisture change for each cylinder can be controlled for each track to simulate temporal decorrelation.

PolSARproSim+ Modifications

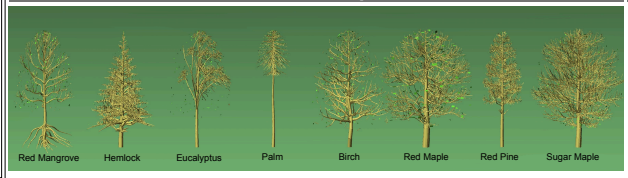
PolSARproSim has been modified to:

- Have increased flexibility in describing the simulated forest by
  - Having control over detailed structural parameters for any number of species defined via an XML database
  - Having any number of species in a simulated forest at once (mixtures)
  - Having control over properties of each tree such as location, height, dbh etc. via a separate XML database.
- Be able to simulate stacks of multi-baseline interferograms in high fidelity (slow) and approximate (fast) modes
- Be able to introduce temporal decorrelation through physical change such as movement of canopy elements (branches and leaves) and moisture change
- Include realistic band-limited Point Spread Functions (raised-cosine weighted) and thermal noise effects
- Be able to simulate returns from arbitrary topography
- Make the code run faster (i.e. multi-threaded implementation)
  - An approximate 5-fold decrease in simulation time for a single baseline configuration

Forest Definition

PolSARproSim+ allows unprecedented control over definition of the forest stand that it simulates. This is accomplished through two XML database files. The first, defines the forest stand. It contains primitives of each and every tree that will be simulated by PolSARproSim+. This allows simulation of radar returns from a stand with any stem density, tree heights and species density desired. Each tree defined in the stand belongs to a particular species. Species specific parameters are defined in the second XML-database file. This database contains an extensive set of parameters giving the user extensive control and the ability to define any number of species.

Allometric Definition of Species



A tree in PolSARproSim+, is a complex object defined by a whole slew of allometric parameters. These parameters control the fine structural details of a particular realization of a tree, allowing flexibility in defining any number of tree species in significant detail, illustrated by a few examples in the figure above. Similar to what is seen in nature, each individual tree is unique, however the allometric parameters defined in the XML database make trees belonging to a species appear similar.

The example on the right shows a snippet of the allometric XML database, where the primary branch parameters of a particular species are defined. The parameters of the stem, secondary and tertiary branches, crowns and foliage are similarly defined. Since the number of parameters needed to define a particular species is quite high, the structure of the species allometry file is designed so the user can define a species without having to specify each and every parameter. This is accomplished by letting each species belong to a general category, say for instance deciduous or coniferous. While all allometric parameters must be defined for a category, parameters for a particular species are adopted from its category unless explicitly specified in the species tag.

```
<primary_polar_angle_mean      units="deg">   50.0        </primary_polar_angle_mean>
<primary_polar_angle_stdev     units="deg">   25.0        </primary_polar_angle_stdev>
<primary_layer_density         units="none">  2.01        </primary_layer_density>
<primary_sections              units="none">  2           </primary_sections>
<primary_azimuth_factor        units="deg">   90          </primary_azimuth_factor>
<primary_radius_quadratic_coeff units="none"> 0.0         </primary_radius_quadratic_coeff>
<primary_radius_linear_coeff   units="none">  -0.3465     </primary_radius_linear_coeff>
<primary_radius_constant_coeff units="none">  0.693      </primary_radius_constant_coeff>
<primary_tropism_factor        units="none">  0.1         </primary_tropism_factor>
<primary_tropism_polar_angle   units="deg">   0.0         </primary_tropism_polar_angle>
<primary_tropism_azimuth_angle units="deg">   0.0         </primary_tropism_azimuth_angle>
<primary_lambacx_stdev         units="none">  0.25        </primary_lambacx_stdev>
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<primary_dry_sections          units="none">  5           </primary_dry_sections>
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<primary_dry_polar_angle_stdev units="deg">   25.0        </primary_dry_polar_angle_stdev>
```
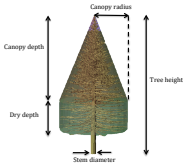

Forest Primitives

The figure on the right illustrates the parameters used to define a tree within a forest definition database. As in a typical XML file, all attributes are defined between XML tags. Each tree is defined by eight primitives; species, x,y location, height, canopy radius, depth of the live and dry canopies and stem diameter (or dbh) specified within their respective XML tags. This allows the user extensive, unprecedented control over the definition of the simulated stand.




```
<tree>
        <species        units="none">      RHMA       </species>
        <x_location     units="meters">    0.8400     </x_location>
        <y_location     units="meters">    1.0000     </y_location>
        <tree_height    units="meters">    18.0900    </tree_height>
        <canopy_radius  units="meters">    3.8770     </canopy_radius>
        <canopy_depth   units="meters">    12.0900    </canopy_depth>
        <dry_depth      units="meters">    5.0141     </dry_depth>
        <stem_diameter  units="meters">    0.2243     </stem_diameter>
</tree>
```

The above example shows a snippet of the forest definition database, where a tree (in this case the species RHMA or 'Red Mangrove') is defined by its primitives; the location, height, diameter and canopy parameters. PolSARproSim+ simulates the first N trees defined in the XML file, with N separately defined in the same file. The user is allowed the flexibility of not having to define each and every tree, instead a few parameters defined in the input parameter file suffice, much like what existed in PolSARproSim.

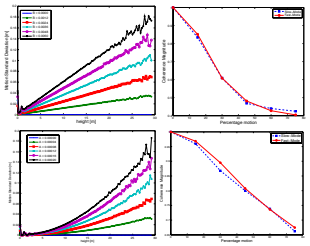
Simulating Temporal Decorrelation

PolSARproSim+ can simulate the effects of temporal decorrelation by randomly moving or changing the moisture of individual scattering elements, for example the cylinders created from sections of the stems and branches. The amount of change in moisture or position is specified by the user for each track as the set of coefficients (e.g. A, B, C) of a quadratic or exponential height profile

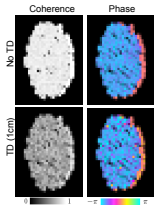
$$\sigma(h) = Ah^2 + Bh + C$$
$$\sigma(h) = Ae^{Bh} + C$$

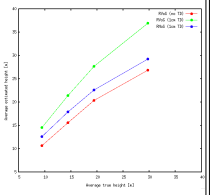
This profile can be specified relative to the ground or the bottom of the canopy. The position of a scattering element (a three dimensional vector from the scene center to the center of the scatterer) and its moisture is changed using the following set of equations

$$M_{new} = M + dM \qquad \vec{p}_{new} = \vec{p} + \vec{d}\vec{p}$$
$$dM \sim N\left(\sigma(h), \sigma^2(h)/F\right) \qquad \vec{d}\vec{p} = \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$
$$dx, dy, dz \sim N\left(0, \sigma^2(h)\right)$$



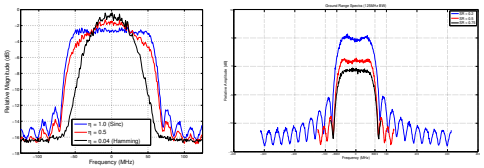
The plots on the left show six linear (top) and quadratic (bottom) position change profiles specified for each track and the plots on the right show the corresponding mean temporal decorrelation observed over a stand



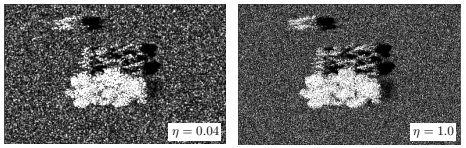


PolSARproSim+ also allows for an analysis of the effects of temporal decorrelation on PolInSAR signatures and inversion algorithms. The figure above shows results of PolInSAR inversions using the three stage RVoG and RMoG [7] algorithms on stands with an average tree height varying from 10 to 30m with a linear temporal decorrelation profile (such that there is 1cm rms motion at 20m height). There is a clear improvement in height inversions using the RMoG model where temporal decorrelation is modeled as a parameter.

Point Spread Function Implementation



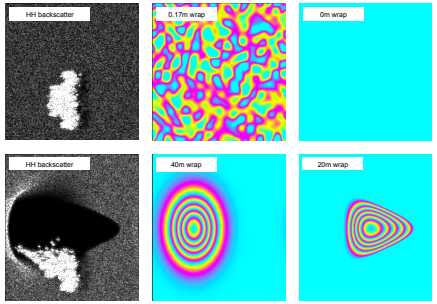
The shape of the point system point spread function can be modified in PolSARproSim+ using the broadening factor η as shown in the range-spectra (on the left) for three different values of η. The sampling of the range and azimuth spectra can also be changed by setting the sampling factor (the ratio of pixel dimension to resolution for range and azimuth individually), as seen on the right for three different values of the sampling



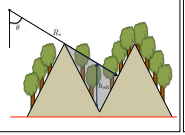
These two images highlight the qualitative difference between two simulated images, on the left with a broadening factor of 0.04 simulating a 'Hamming-weighted' point-spread function and on the right with a broadening factor of 1.0 simulating a 'Sinc' point-spread function, calculated using the following equation:

$$I(\delta_x) = \mathrm{sinc}\left(\frac{\pi\delta_x}{\Delta x}\right) + \frac{1}{\pi}\frac{1-\eta}{1+\eta}\sin\left(\frac{\pi\delta_x}{\Delta x}\right)\left(\frac{\delta_x/\Delta x}{1-(\delta_x/\Delta x)^2}\right)$$

Simulation using an arbitrary DEM



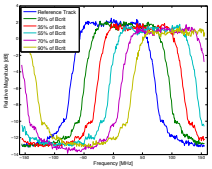
PolSARproSim+ allows simulation of a forest over any arbitrary terrain supplied through an external DEM. Once read the DEM is resampled to the simulation coordinates using a simple bilinear interpolation. In order to account for scatterers occluded from the radar because of large slopes, a map of the shadow heights is then generated using geometry from the master track. This 2-D map contains the height up to which an object in that particular location would be hidden from the radar. The top three images show simulation without a DEM, and the bottom three show the same simulation, with an external DEM. Image to the right illustrates shadow height map

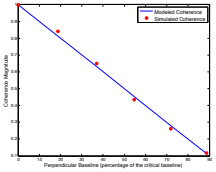


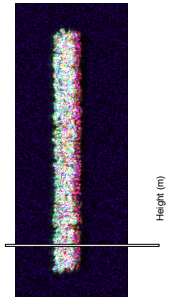
Multibaseline Simulations

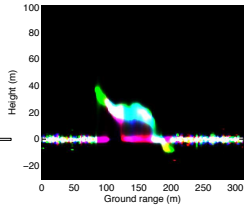
PolSARproSim+ allows simulation of multibaseline data. Track geometry is defined in the parameter file as cross-track and vertical baseline offsets from the master track. The two figures here show the range spectra for scenes simulated with varying baselines chosen as a function of the critical baseline (i.e. the baseline where geometric decorrelation is at a maximum) and the corresponding coherence for interferograms formed using these scenes. The observed spectral shift corresponds to the predicted Prati shift while the the amount of decorrelation follows the geometric decorrelation model given by

$$\gamma_G = 1 - \frac{B_\perp}{B_{\perp c}}$$

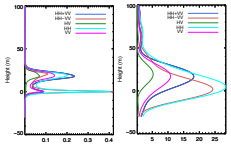








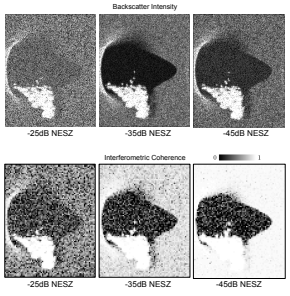
Transects of a tomogram from a 7-track simulation of an exact stem map shows the ability of PolSARproSim+ to simulate realistic multibaseline data from conditions very similar to those seen in reality. The scene simulated in the above example contains roughly 1200 trees along a 200m by 50 transect in the Harvard Forest from a stem-map created by the University of Massachusetts, Amherst researchers that contains x-y locations, species, and diameter data for each tree. These attributes are easily imported into the simulator by design.



Thermal Noise Effects

PolSARproSim+ allows the simulation of thermal noise through the parameter 'NESZ', the noise equivalent sigma naught, supplied in dB for each polarization channel. The simulated scattered field with thermal noise is then given by the following set of equations, where the subscript 'pq' denotes a particular polarimetric channel. The noise is uncorrelated between different tracks, which causes decorrelation in interferometric coherence as seen here

$$S_{N_{pq}} = S_{pq} + N_{pq}$$
$$N_{pq} = n_{x_{pq}} + in_{y_{pq}}$$
$$n_{x_{pq}}, n_{y_{pq}} \sim N\left(0, \frac{\sigma_n^0}{\sqrt{2}}\right)$$
$$\sigma_n^0 = 10^{(-NESZ_{ch}/20)}$$



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